GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code: DI05023061

Subject Name: Designing of Blow and Thermoforming Moulds

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PEC-04

Prerequisite:	None
Rationale:	It is very interesting to find numerous products in the area of packaging of soft drink, mineral water bottle, pharmaceutical and cosmetic items which are made by blow molding and thermoforming process. Hence, it is desirable to have competencies in designing and developing variety of mold for the same. A Plastic Diploma engineer has to monitor operations and maintenance of Blow and Thermoforming mold. This competency requires the knowledge of constructional details of Blow and Thermoforming Molds. Hence the course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Select suitable blow mold material.	R,U
02	Design blow molds for various products.	R,U,A
03	Design efficient cooling system for blow mould.	R,U,A
04	Select suitable thermoforming mold material.	R,U,A
05	Design thermoforming molds for various thermoformed products.	R,U,A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Blow mould Materials	• Introduction of Blow Molds • Mould Material Selection Requirements • Types of Mould Materials • List of materials • Advantages & disadvantages of materials	5	10
UNIT – II Blow Mould Design Considerations	2.1 Injection Blow Mould • Mould Construction, Injection Blow Mould Design, Core rods / Parison Stick Design 2.2 Extrusion Blow Mould • Design procedure of Extrusion Blow Mould. • Mould Cavity Design • Method for deciding clamping force • Design consideration for mould parting line • Mould finish and Mould Venting • Cavity surface considerations • Welding edges & flash pockets in pinch-off • Neck insert design considerations • Radii and corner edge designs • Molded holes and handle design • Shrinkage considerations • Blow and swell ratio • Molds for undercut products (moving section molds) • Flash removal methods • Mould cleaning methods	14	30
UNIT – III Blow Mould Cooling and Ancillary Elements	3.1 Mould Cooling • Significance of mould cooling. • Cooling designs for neck, pinch-off and main body. 3.2 Ancillary Elements • Base plates • Alignment pins • Striker plates • Ejection Methods	07	20
UNIT – IV Thermoforming Mould Materials	4.1 Thermoforming Mould Materials • Introduction of Thermoforming Moulds • Mould Material requirements	05	10

	• Types of mould materials for prototype and production tooling		
UNIT – V Thermoforming Mould Design Considerations	5.1 Vent Holes/Vacuum Holes: Number of vents, Size, Position and types 5.2 Shrinkage and Draft Angle Considerations 5.3 Plug Design (For Plug Assist Mould)- Plug material, Shape of plug, Plug design concepts 5.4 Sheet clamping mechanisms 5.5 Draw ratios and its importance 5.6 Chamfers and radii 5.7 Surface treatments 5.8 Surface texture 5.9 Mould cooling designs 5.10 Use of moving elements - collapsing cores, Unscrewing devices, cammed sections and slides 5.11 Ejection techniques 5.12 Application of Mould Releasing Agents 5.13 Moulds for undercuts 5.14 Multi-impression Moulds • Factors to be considered for deciding correct impression layout • Method of determining the number of impressions for a given forming platform size and product size • Design and draw thermoforming mould for multi-impressions.	14	30
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
25	**50**	**25**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S.No.	Title of Book	Author	Publication with place, year and ISBN
1	Blow Moulding Handbook	Rosato/Rosato	Van Nostrand Reinhold, 1989, 9783446150713
2	Plastic Blow Moulding Handbook	Norman Lee	Van Nostrand Reinhold, 1990, 9789401169905
3	Blow Moulding	Fisher	Butterworth & Co.,1971, 9780592054384
4	Technology of Thermoforming	James L. Throne	Hanser Gardner Publications, 1996, 9781569901984
5	Fundamentals of Plastics Thermoforming	Peter Klein	Morgan & Claypo Publishers, 2009, 9781598298840

(b) Open-source software and website:

1. https://geminigroup.net/resources/blow-mold-design-guide/
2. http://www.custom-pak.com/design/blow-molding-design-guide/
3. https://techcenter.lanxess.com/scp/americas/en/docguard/Part_and_Mould_Design_Guide.pdf?docId=77015
4. http://www.plastiglas.com.mx/images/content/PLASTIGLAS_INST/uploads/1168103546998Thermoforming.pdf
5. http://www.multifab-inc.com/guidelines.pdf
6. https://multifabmanufacturing.com/assets/Uploads/Multifab-Thermoforming-Guidelines-Rev-3-12-18.pdf
7. https://kompozit.org.tr/wp-content/uploads/2021/11/A_Design_Guide_Part_and_Mold_Design_Engi.pdf
8. https://www.youtube.com/watch?v=N3FXPDx4_3c
9. https://www.youtube.com/watch?v=30MD5teExn8

Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	Design and draw sectional elevation, plan and inverted plan of blow moulds. (For different shapes to be casted, number of sheets would depend upon complexity of shapes. Faculty should give enough number of sheets to students to justify the time allocated)	14
2	Draw detail drawings of mould drawn above.	08
3	Design and draw sectional elevation, plan and inverted plan of Thermoforming mould. (For different shapes to be casted, number of sheets would depend upon complexity of shapes. Faculty should give enough number of sheets to students to justify the time allocated)	08
TOTAL		**30**

Suggested Activities for Students:

1. Students will select material for blow mould.
2. Students will design blow mould.
3. Students will collect various blow molded products
4. Students will design thermoform mold
5. Students will visit nearby tool room related industry.

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